Team Signum



- 📌 **Model Summary**
- **Baseline Model** Logistic Regression was used as a benchmark. It provided interpretability but lower performance compared to advanced models.
- **Main Model** LightGBM was selected as the final model due to superior performance on AUC, Average Precision, and Precision@K.
  - Tuned with **Optuna Bayesian optimization**.
  - Key hyperparameters: num_leaves, max_depth, learning_rate, n_estimators, subsample, colsample_bytree.
- **Evaluation Metrics** Weighted scoring:
  - AUC-ROC (40%)
  - Average Precision (25%)
  - Precision@10% (20%)
  - Recall@10% (15%) → LightGBM consistently outperformed baseline across all metrics.
- **Explainability** SHAP analysis identified top churn drivers:
  - **freq_30d** (transaction frequency)
  - **balance trend** (declining balances)
  - **service diversity** (number of products used)
  - **recency** (time since last activity)
  - **peer diversity** (network connections)
- **Business Linkage** Each SHAP driver was tied to actionable interventions (notifications, rewards, bundles, win-back campaigns, referrals). → Ensures model outputs translate directly into retention strategies.



- Each recommendation is tied to a driver identified by SHAP:
- **freq_30d (transaction frequency)** → Send re-engagement push notifications to low-frequency users 2 weeks before predicted churn.
- **balance trend (declining balances)** → Offer loyalty rewards or cashback to stabilize account activity.
- **service diversity (few product types used)** → Promote bundle offers or cross-sell campaigns to increase engagement.
- **recency (long inactivity gaps)** → Launch win-back campaigns (emails, SMS) to re-activate dormant users.
- **peer diversity (limited network connections)** → Encourage referral programs to expand peer activity.



- **Customer Handling Strategy**
- **Low-frequency users** *Driver: freq_30d* → Send **re-engagement push notifications** about offers or reminders **2 weeks before predicted churn**.
- **Declining balance customers** *Driver: balance trend* → Provide **cashback, loyalty rewards, or fee waivers** to stabilize account usage.
- **Single-service users** *Driver: service diversity* → Promote **bundle offers or cross-sell campaigns** to increase product stickiness.
- **Inactive customers** *Driver: recency* → Trigger **personalized win-back campaigns** (emails, SMS, calls) after **21 days of inactivity**.
- **Low peer diversity customers** *Driver: peer diversity* → Encourage **referral programs** and **social engagement features** to connect them with peers.



THANK YOU